

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-404 — Microprocessor

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks**Question 1.** Explain the pin diagram of the 8085 microprocessor in detail.**Question 2.** Describe the internal block diagram of the 8085 microprocessor.**Question 3.** What are interrupts? Differentiate between software interrupts and hardware interrupts in the 8085 microprocessor.**Question 4.** How many types of addressing modes does the 8085 microprocessor have? Explain any two of them with suitable examples.**SECTION — B****Question 5.** Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) Explain the three-step instruction execution cycle (Fetch, Decode, Execute) in detail.
- (b) What are universal logic gates? Explain why they are called so.
- (c) Differentiate between a microprocessor, a microcontroller, and a microcomputer.
- (d) What is assembly language? Discuss its advantages over machine language.
- (e) Explain the computer memory hierarchy diagram in detail.
- (f) Explain all three system buses: address bus, data bus, and control bus.
- (g) Describe the instruction format and instruction types of the 8085 microprocessor.
- (h) Draw a block diagram representing a microprocessor-based system and explain its basic components and functionality.
- (i) Explain the architectural block diagram of the 8085 microprocessor.

SECTION — C**Question 6.** Attempt all parts:**10 × 1 = 10 Marks**

- (a) What is a microprocessor?

- (b) Write the list of logical operations performed by the 8085 microprocessor.
- (c) What are the three main units that a microprocessor-based system is made of?
- (d) Convert binary number “101101110” into its decimal equivalent.
- (e) Draw the logic gate symbol that represents a NAND gate.
- (f) Find out the 2’s complement of the binary number “001001110”.
- (g) What is machine language?
- (h) Is the pin diagram of the 8085 microprocessor considered an external diagram or an internal diagram?
- (i) Write down the functionality of the $\overline{RD}$ and $\overline{WR}$ signals in brief.
- (j) What are maskable interrupts in the 8085 microprocessor?